

Simulating viral disease using Monte Carlo simulations

Physics equations: -

The continuous behaviour of the epidemic is governed by the following differential rate equations:

$$dS/dt = -k_2SI$$

$$dI/dt = k_2SI - k_1I$$

$$dR/dt = k_1I$$

I have translated these continuous macroscopic equations into a discrete, microscopic simulation. In our code, k_2 represents the transmission rate which indicates the daily probability an infected student spreads the virus to a susceptible one and k_1 represents the recovery rate which represents the 20% daily chance an infected student moves into the removed category.

How I solved the problem: -

1. Initialization: I created a 1D NumPy array representing 1,000 students. I assigned states of the students based on integers: 1 (Susceptible), 2 (Infected), and 0 (Immune/Removed). Finally, I randomly assigned the initial 1% immunity and the first patient as zero.
2. Time Progression: I developed a loop to step through 40 individual days.
3. Interaction & Transmission: On each day, the loop checks every student. If they are infected, the code selected another student at random. If that target is susceptible, the simulation change their state to infected.
4. Recovery: For every infected student, I used the code to generate a random number between 0 and 1. If it falls below 0.2, the code changed their state to immune.
5. Averaging: Because human interaction is random, a single 40-day run is noisy. I wrapped the entire simulation in an outer loop to repeat the 40-day timeline 40 separate times, averaging the arrays to produce a reliable data.